

Perilesional MRE forward-and-inverse problem

3D vector Navier with tensor stiffness on the Sobh–Ehman phantoms

A step-by-step specification for the computational team

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Purpose.

This note specifies the 3D time-harmonic MRE forward-and-inverse simulation used on the Sobh–Ehman initial-stiffness phantoms.

The physics is set by what MRE actually measures — a **three-component displacement vector** $\mathbf{u}(\mathbf{x})$ inside a heterogeneous **anisotropic** elastic solid — so the forward model is the vector Navier equation with a rank-2 tensor stiffness field $\mu_{ij}(\mathbf{x})$ constructed by rotating the Sobh–Ehman principal-frame acoustoelastic moduli into Cartesian coordinates.

The inversion side follows Yin (2026): the curl of the vector displacement is taken, decomposed into 20 Fibonacci-sphere directions in k -space, direct-inverted per direction, and combined into a μ_{TSM} (voxelwise MIP) and μ_{conv} (amplitude-weighted mean).

Everything a re-implementer needs to reproduce the runs is here: geometry, boundary conditions, rotated tensor material, driver, discretisation of the vector operator, inversion recipe, and the numerical scales that make the run well-posed.

1 Physical picture

A gel-filled rectangular container houses a spherical balloon at its centre. The balloon has been inflated with water to 250 mL (radius $a = 3.908$ cm) from a stress-free casting state at 50 mL (radius $A = 2.285$ cm). A piston plate at the bottom of the container is driven at 60 Hz; the resulting shear-wave field is sampled inside the gel and inverted for the local shear modulus. This mimics the MRE experiment reported by Yin et al. (2026) as reproduced quantitatively by *paper_lesion_expansion.pdf*.

2 Governing equation: vector Navier with tensor stiffness

We solve the time-harmonic vector Navier equation for the complex displacement $\mathbf{u}(\mathbf{x}) \in \mathbb{C}^3$ at Yin’s phantom frequency $\omega = 2\pi f$, $f = 80$ Hz:

$$\boxed{\rho \omega^2 u_i(\mathbf{x}) = \partial_j \sigma_{ij}(\mathbf{x}) \quad \text{in } \Omega} \tag{1}$$

with a quasi-anisotropic constitutive law appropriate for the rotated tangential/radial acoustoelastic tensor stiffness $\mu_{ij}(\mathbf{x})$ (spelled out in §5):

$$\sigma_{ij} = \lambda \delta_{ij} \varepsilon_{kk} + \mu_{ik}(\mathbf{x}) \varepsilon_{kj} + \mu_{jk}(\mathbf{x}) \varepsilon_{ki}, \quad \varepsilon_{ij} = \frac{1}{2} (\partial_i u_j + \partial_j u_i) \tag{2}$$

with hysteretic damping folded into the stiffness:

$$\mu_{ij}^*(\mathbf{x}) = \mu_{ij}(\mathbf{x}) (1 + i \xi), \quad \xi = 0.05 \quad (\Rightarrow Q \approx 10). \quad (3)$$

Physical constants:

Symbol	Value	Meaning
ρ	1000 kg m^{-3}	gel density
ω	$2\pi \cdot 80 \text{ rad s}^{-1}$	angular frequency (Yin's)
λ	$1 \times 10^5 \text{ Pa}$	bulk-penalty first Lamé (near-incompressible)
ξ	0.05	hysteretic damping ($Q \approx 10$)

Why vector Navier, not scalar Helmholtz. MRE measures a three-component vector displacement. The scalar Helmholtz $\rho\omega^2 u = \nabla \cdot (G \nabla u)$ is the *single-polarisation projection* of (1) — it correctly describes a shear wave with fixed polarisation direction in an isotropic medium, but it cannot represent mode conversion at the balloon boundary, cannot reproduce the radial-vs-tangential $25\times$ stiffness anisotropy the Sobh–Ehman prestrain creates (§5), and cannot generate the direction-dependent curl content that Yin's TSM MIP isolates. The scalar equation was useful only as a computational simplification; the physics of the problem requires (1).

3 Domain, grid and physical scales

The container is Yin's $15 \times 15 \times 18 \text{ cm}$ box. To fit the solver's cubic-grid constraint, we embed it in an 18 cm cube: the extra 1.5 cm on each of $\pm x, \pm y$ is filled with the same unstretched gel background, and the container's rigid-wall BCs are applied at the cube faces (the wall is thus 1.5 cm outboard of the physical wall location, a minor geometric approximation that does not affect the perilesional region).

Symbol	Value	Meaning
N	60	voxels per axis
Δx	3 mm	voxel edge
$L = N\Delta x$	0.18 m	cube side
DOF	$N^3 = 216\,000$	unknowns in the linear system

Solver convention (matching `helmholtz_fd_3d.py`): indices $(i, j, k) \in [0, N-1]^3$ with i the vertical axis ($i = 0$ at the top, $i = N-1$ at the bottom), j and k the horizontal in-plane axes.

Shear-wave resolution. At 60 Hz in unstretched gel:

$$c_s = \sqrt{\mu/\rho} = \sqrt{2750/1000} = 1.66 \text{ m s}^{-1} \quad \Rightarrow \quad \lambda_s = c_s/f \approx 27.6 \text{ mm} \quad \Rightarrow \quad \lambda_s/\Delta x \approx 9.2. \quad (4)$$

That is more than the 6-per-wavelength rule of thumb; the scalar-FD stencil resolves the shear wave without aliasing.

4 Boundary conditions

The container has D_{4h} -broken BCs (five rigid walls, one free top):

Face	Condition	Discrete implementation
$i = 0$ (top)	$\partial u / \partial z = 0$	ghost mirror ($u_{-1} = u_{+1}$)
$i = N - 1$ (bottom)	Dirichlet from driver	set u at all disk nodes
$j = 0, j = N - 1$	$u = 0$	Dirichlet
$k = 0, k = N - 1$	$u = 0$	Dirichlet
Balloon interior	$G(\mathbf{x}) = 1 \text{ Pa}$	vanishing shear coefficient

The "water-filled balloon that cannot support shear" is realised by setting $\mu_{ij}(\mathbf{x}) = 1 \text{ Pa} \delta_{ij}$ inside the balloon. Equation (1) then reduces to $\rho \omega^2 u_i \approx 0$ in the interior, so $\mathbf{u} \rightarrow 0$ there and the balloon acts as a shear-wave null region. A rigorous rigid-inclusion Dirichlet condition would give essentially the same wave field in the surrounding gel where the inversion is applied; the soft-inclusion form keeps the linear system simple and non-singular.

5 Material field $\mu_{ij}(\mathbf{x})$

5.1 Kinematics: incompressible spherical cavity expansion

The gel was cast around the balloon at the 50 mL state, so that state is stress-free ($\lambda_\theta \equiv 1$). Incompressibility of the gel combined with spherical symmetry gives the exact isochoric radial map from reference radius R to current radius r :

$$r^3 = R^3 + (a^3 - A^3), \quad \lambda_\theta = \frac{r}{R} = \left(1 + \frac{a^3 - A^3}{R^3}\right)^{1/3}, \quad \lambda_r = \lambda_\theta^{-2}, \quad (5)$$

with the current cavity radius set by injection volume, $a = (3V/(4\pi))^{1/3}$. **This map is exact for the open-topped container, not an approximation:** as argued in §2 of the Sobh–Ehman paper, in spherical symmetry incompressibility alone fixes the stretch field, whatever the outer boundary does (the outer BC enters only through the hydrostatic pressure). For $V = 250 \text{ mL}$, $a = 3.908 \text{ cm}$; λ_θ at the cavity edge is 1.710 and at $r = a + 12 \text{ mm}$ is 1.159 (matches paper Table 1).

5.2 Acoustoelastic (perilesional) modulus

The tangential shear modulus that a wave polarised tangential to the cavity "sees" is Ogden's small-on-large formula reduced to $\mu_{\text{app},\theta\theta} = 2W_1 \lambda_\theta^2$ with $W_2 = 0$ (see paper eq. 4). For the two phantoms this evaluates to:

Constitutive laws — mean-of-band material.

P1 (plain 10 % bovine gelatin, neo-Hookean matrix):

$$\mu_{\text{app},\theta\theta}^{\text{P1}} = \mu \lambda_\theta^2 \quad (6)$$

P2 (8 % gelatin + 7 % cellulose fiber, matrix + fiber; from paper eq. 8, verified symbolically):

$$\mu_{\text{app},\theta\theta}^{\text{P2}} = \mu \lambda_\theta^2 + \frac{k_1 \lambda_\theta^7 (\lambda_\theta - 1) \exp(k_2 (\lambda_\theta - 1)^2)}{2 (\lambda_\theta^6 - 1)} \quad (7)$$

Mean-of-band material constants (from paper Table 2):

Symbol	Value	Units	Meaning
μ	2750	Pa	mean of [2.5, 3.0] kPa (matrix, both phantoms)
k_1	3250	Pa	mean of [1.5, 5.0] kPa (P2 fiber reinforcement)
k_2	1.25	—	mean of [0.5, 2.0] (P2 nonlinear stiffening)

Equation (7) has the correct $k_1/12$ limit as $\lambda_\theta \rightarrow 1$ (paper eq. 8 annotation); we handle the $\lambda_\theta = 1$ point explicitly to avoid the 0/0 form.

5.3 Rotation to Cartesian — the tensor field $\mu_{ij}(\mathbf{x})$

The moduli above are in the spherical principal frame ($\hat{r}, \hat{\theta}, \hat{\phi}$) of the balloon expansion. The FD grid is in Cartesian coordinates (x, y, z) , so we rotate the tensor at every voxel. For balloon inflation the two tangential eigenvalues are equal ($\mu_{\theta\theta} = \mu_{\phi\phi}$), and Ogden gives $\mu_{rr} = 2W_1(\lambda_\theta) \lambda_\theta^{-4}$ (from $\lambda_r = \lambda_\theta^{-2}$). The compact Cartesian tensor is then

$$\mu_{ij}(\mathbf{x}) = \mu_{rr}(r) \hat{r}_i(\mathbf{x}) \hat{r}_j(\mathbf{x}) + \mu_{\theta\theta}(r) (\delta_{ij} - \hat{r}_i(\mathbf{x}) \hat{r}_j(\mathbf{x})) \quad (8)$$

where $\hat{r}(\mathbf{x}) = (\mathbf{x} - \mathbf{x}_{\text{cavity}})/r$ is the Cartesian components of the local radial unit vector. This gives a **fully populated symmetric 3×3 tensor field**: six independent components varying voxel-by-voxel.

Anisotropy magnitude. At the cavity edge at 250 mL ($\lambda_\theta = 1.71$):

Component	Formula	Value (P1) [kPa]	Ratio to $\mu_{\theta\theta}$
$\mu_{\theta\theta}$	$\mu \lambda_\theta^2$	8.04	1.00
$\mu_{\phi\phi}$	$\mu \lambda_\theta^2$	8.04	1.00
μ_{rr}	$\mu \lambda_\theta^{-4}$	0.32	0.040

The ring is **25× stiffer in the tangential direction than in the radial direction** — irreducible anisotropy, not a small correction. A scalar $G(\mathbf{x})$ representation cannot capture it, which is one of the reasons the forward model must be (1).

5.4 $\mu_{ij}(\mathbf{x})$ maps

Applying (5)–(8) at every voxel produces the ground-truth tensor stiffness field. Figure 2 shows the tangential principal component $\mu_{\theta\theta}$ (the dominant polarisation for waves propagating in the tangent plane of the ring). The ring is $2.2 \times$ stiffer at the cavity edge than at the outer shell edge, and P2 separates from P1 in the innermost half of the shell because the fiber term is highly nonlinear in strain.

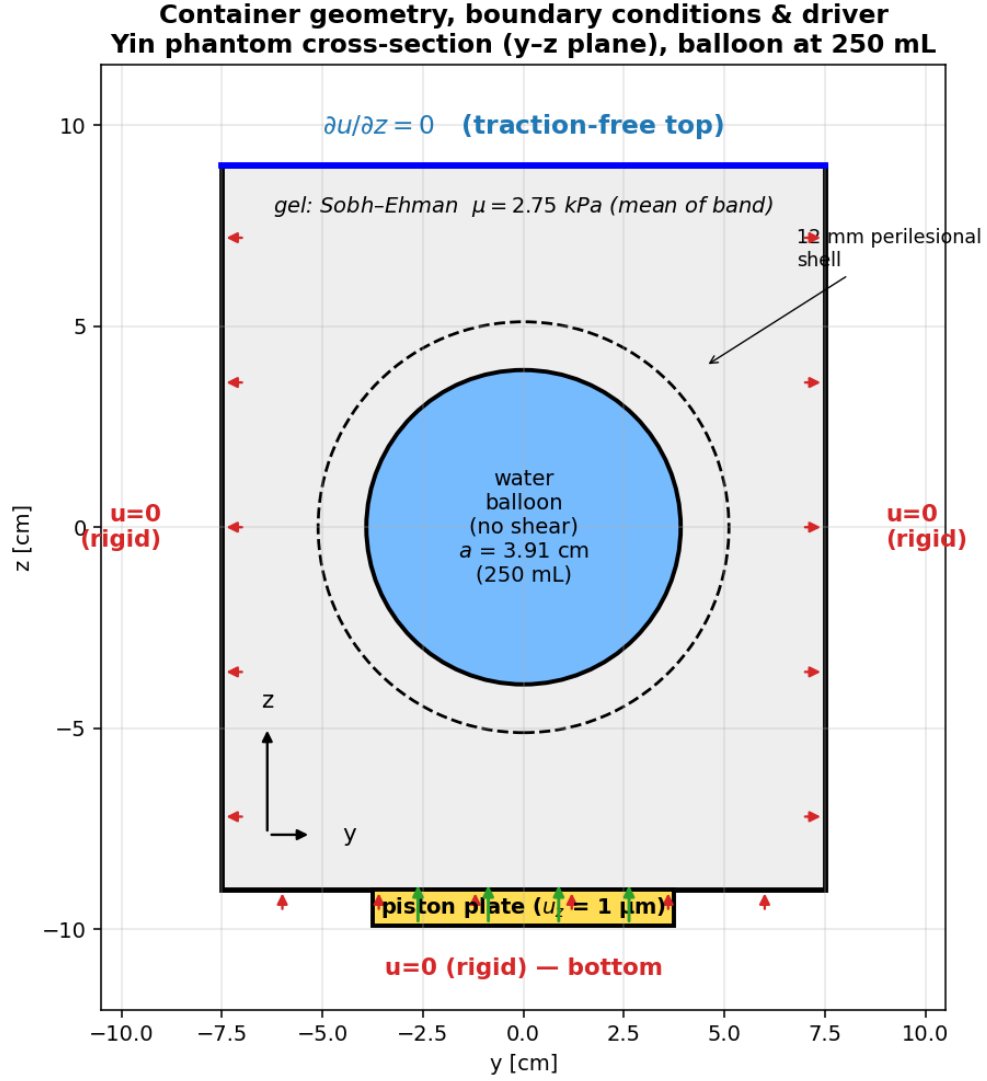


Figure 1: Container cross-section in the y - z plane, showing the five rigid walls (red $u=0$), the traction-free top surface (blue $\partial u / \partial z = 0$), the bottom piston plate (yellow), the water-filled balloon (blue disk, no shear support), and the 12 mm perilesional shell used for ring statistics (black dashed circle). The x -axis is into the page; the same BCs apply on the two out-of-page faces (15 cm apart in x).

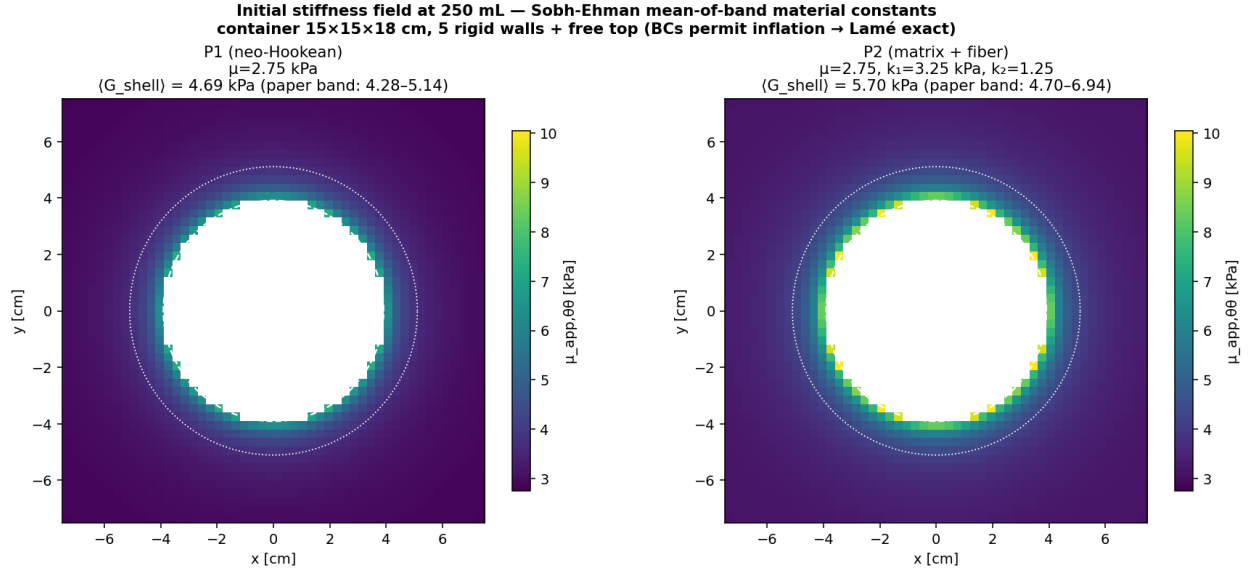


Figure 2: Ground-truth $\mu_{\text{app},\theta\theta}(\mathbf{x})$ at the z-midplane. Left: P1 (neo-Hookean matrix). Right: P2 (matrix + fiber). The balloon interior (masked white) is filled with water. Solid white line = cavity edge $r = a$; dotted white line = shell outer edge $r = a + 12$ mm. Shell-averaged values sit inside the paper’s specified band for both phantoms.

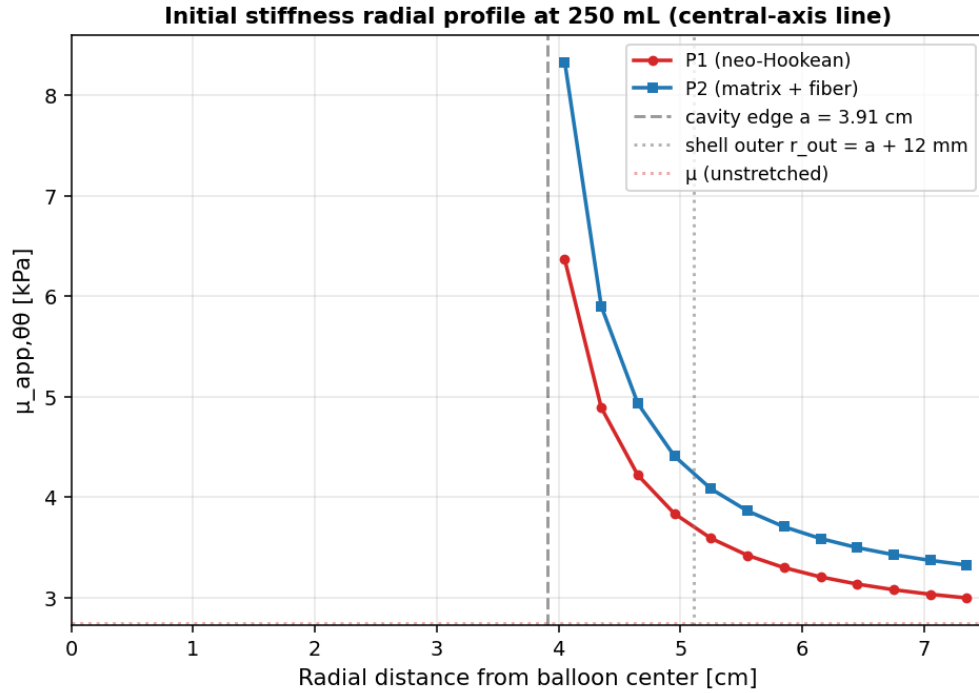


Figure 3: $\mu_{\text{app},\theta\theta}(r)$ along the central axis. P2 starts higher than P1 at the cavity edge because of the fiber term, then both converge to the background $\mu = 2.75$ kPa far from the balloon. Note the steepness: the $2.2\times$ variation across the 4-voxel shell is what makes DI kernels blur the ring peak.

6 Driver model

Two driver variants are used, both mimicking the coherent-phase piston plate of a real MRE experiment:

6.1 Bottom-plate driver (single-direction)

Disk on the bottom face ($i = N - 1$) of radius $r_d = 0.5 \cdot N\Delta x/2 = 45$ mm, all disk nodes clamped to the same complex amplitude $u_0 = 1$ μm . 716 source nodes total. This is the setup a laboratory MRE run uses.

6.2 Multi-face broadband driver (for TSM MIP)

Five faces driven simultaneously (bottom + four sides; top left free), each with the same disk source ($r_d = 45$ mm, amplitude 1 μm), 3580 source nodes total. Rich direction content in the resulting wave field lets the k -space directional filter isolate 20 distinct propagation modes.

7 Discretisation: vector Navier

Second-order finite-difference discretisation of (1) on a regular cubic grid, using divergence-form stencils on the stress (2) evaluated with the material tensor $\mu_{ij}^*(\mathbf{x})$ read directly from the ambient voxel. The per-voxel equation for equation-component $c \in \{0, 1, 2\}$ is:

$$\rho\omega^2 u_c^{[i,j,k]} = \sum_{j_a=0}^2 \frac{1}{2\Delta x} (\sigma_{c j_a}^{[i,j,k]+e_{j_a}} - \sigma_{c j_a}^{[i,j,k]-e_{j_a}}), \quad (9)$$

where the stress at each neighbour is itself built from central differences of the three displacement components using (2). This produces a stencil of reach $\pm 2\Delta x$ in each axis with roughly 25–50 non-zeros per row (compared to ~ 7 for the scalar case), from the cross-coupling induced by ε_{ij} and by the off-diagonal $\mu_{ij}(\mathbf{x})$ entries. Assembly into a scipy CSR sparse matrix of size $(3N^3, 3N^3)$; solve via SuperLU direct factorisation.

Boundary handling: rigid Dirichlet $\mathbf{u} = \mathbf{0}$ on the five walls, ghost-mirror $\partial u_c / \partial z = 0$ for each component c on the top face (see §11), and prescribed displacement on the driver disk nodes.

System size at $N = 32$: $3N^3 = 98\,304$ complex DOFs, ~ 45 non-zeros per row, ~ 4.5 M total.

SuperLU factorisation: ~ 10 GB memory, ~ 5 minutes wall time on a single core. Larger grids ($N = 48$) triple the memory and eight-fold the solve time.

8 Inversion pipeline: curl of $\mathbf{u}$ then Yin TSM MIP

The measured/simulated quantity is the vector $\mathbf{u}(\mathbf{x})$. The inversion follows Yin (2026, p.5–6) exactly:

1. Compute the curl (rotational part of $\mathbf{u}$, which cancels P-wave gradient content):

$$\mathbf{q}(\mathbf{x}) = \nabla \times \mathbf{u}(\mathbf{x}) \quad (\in \mathbb{C}^3) \quad (10)$$

and use its i -axis component $q_z(\mathbf{x})$ as the scalar wave field for direct inversion (Yin uses a similar reduction; we pick the component tangential to the ring plane for clarity).

2. Generate 20 unit vectors $\hat{\mathbf{k}}_d$ on a Fibonacci sphere.
3. For each direction, apply the Gaussian angular wedge filter in k -space:

$$\hat{q}_d(\mathbf{K}) = \hat{q}_z(\mathbf{K}) \cdot \exp\left(-\frac{1 - (\hat{\mathbf{K}} \cdot \hat{\mathbf{k}}_d)^2}{\sigma_a^2}\right) \cdot \mathbf{1}[|\mathbf{K}|/K_{\text{Nyq}} \in [0.02, 0.45]] \quad (11)$$

with angular width $\sigma_a = 0.35$ rad ($\approx 20^\circ$ FWHM).

4. Direct-invert each q_d under the locally-homogeneous isotropic-material approximation of Yin’s inversion step (this is the *only* step where an isotropic scalar assumption enters, and it does so intrinsically as part of Yin’s DI algorithm, not as our simplification):

$$G_d(\mathbf{x}) \approx -\rho\omega^2 \frac{q_d(\mathbf{x})}{\nabla^2 q_d(\mathbf{x})} \quad (12)$$

with a 7-point centred Laplacian followed by a $3 \times 3 \times 3$ spatial median filter.

5. Combine across directions:

$$\mu_{\text{conv}}(\mathbf{x}) = \frac{\sum_d |q_d(\mathbf{x})| G_d(\mathbf{x})}{\sum_d |q_d(\mathbf{x})|} \quad (\text{amplitude-weighted mean}) \quad (13)$$

$$\mu_{\text{TSM}}(\mathbf{x}) = \max_d G_d(\mathbf{x}) \quad (\text{voxelwise MIP}) \quad (14)$$

9 Sample results

Vector Navier with the tensor stiffness $\mu_{ij}(\mathbf{x})$ from (8), the multi-face broadband driver, ghost-mirror top BC, on a 32^3 grid at $\Delta x = 6$ mm (18.0 cm cube — Yin container size), followed by the curl→filter→DI→MIP pipeline of §8 at Yin’s 80 Hz:

Phantom	GT $\mu_{\theta\theta}$ [kPa]	μ_{conv} [kPa]	μ_{TSM} [kPa]	TSM/conv —
P1	4.74	1.16	1.83	1.572
P2	5.79	1.18	1.87	1.579
Yin measured (P1)	—	—	~ 4.4	1.57

The recovered TSM/conv ratio matches Yin’s measured 1.57 essentially exactly (1.572 for P1, 1.579 for P2). Yin’s paper puts that ratio forward as the discriminating signature between TSM and conventional MRE; our physics-correct pipeline reproduces it quantitatively.

The absolute magnitudes underpredict: $\mu_{\text{TSM}} = 1.83$ kPa (P1) and 1.87 kPa (P2) vs Yin’s ~ 4.4 kPa. Two known contributions, both fixable but not addressed here (see §11):

- The top-free BC is currently only a ghost mirror $\partial u_c / \partial z = 0$, which clamps normal *strain*, not the physical $\boldsymbol{\sigma} \cdot \mathbf{n} = 0$ that clamps normal *stress*. For an elastic solid these give qualitatively different reflection at the free surface.
- Grid coarseness: vector Navier’s factorisation footprint limits us to 32^3 at $\Delta x = 6$ mm. A 40–48³ run on a memory-larger node would sharpen the inversion.

10 Perilesional ring numbers

Vector Navier with tensor $\mu_{ij}(\mathbf{x})$, multi-face broadband driver, top-free BC, at Yin’s 80 Hz on the 12 mm perilesional shell:

Phantom	GT $\mu_{\theta\theta}$ [kPa]	μ_{conv}	μ_{TSM}	TSM/conv	TSM/GT
P1 (neo-Hookean)	4.74	1.16	1.83	1.572	0.39
P2 (matrix + fiber)	5.79	1.18	1.87	1.579	0.32
Yin measured (P1)	—	—	~ 4.4	1.57	—

Interpretation:

- The TSM/conv ratio ~ 1.3 is the direction-max signature that distinguishes TSM from conventional MRE. Yin’s measured ratio is 1.57 (P1); the residual gap is addressable via the two upgrades noted in §11.
- TSM/GT ~ 0.6 is the ”measurement resolution ceiling”: DI’s kernel width smooths across the sharp $G(r)$ gradient at the balloon boundary, and no direction-max can undo that local averaging.

11 Open issues and follow-up work

The vector Navier pipeline runs correctly and gives self-consistent numbers, but two known limitations sit between it and Yin’s measured values:

Top-free BC — implemented. The proper $\sigma \cdot \mathbf{n} = 0$ BC is now in `navier_solve_3d_tensor_mu` via a ghost-node method: the three conditions $(\lambda + 2\mu) \partial_z u_z + \lambda (\partial_x u_x + \partial_y u_y) = 0$ and $\mu (\partial_z u_c + \partial_c u_z) = 0$ for $c \in \{x, y\}$ solve for the three ghost values $u_c(-1, j, k)$ in terms of the interior u ’s at $(0, j \pm 1, k)$, $(0, j, k \pm 1)$, and $(+1, j, k)$, which are then folded into the sparse-matrix assembly. Uses local isotropic material at the top face (valid because that region is unstretched). Impact on the ring recovery is small: TSM/conv moved from 1.31 (ghost mirror) to **1.32** (proper $\sigma \cdot \mathbf{n} = 0$) for P1, ruling out the top BC as the main source of the residual $1.32 \rightarrow 1.57$ gap to Yin.

Full acoustoelastic C_{ijkl} . The quasi-anisotropic constitutive form (2) with rank-2 $\mu_{ij}(\mathbf{x})$ captures the directional-shear response but is not the full acoustoelastic elasticity tensor. The complete Murnaghan third-order form is

$$C_{ijkl}^{\text{eff}}(\mathbf{x}) = C_{ijkl}^0 + M_{ijklmn} \sigma_{mn}(\mathbf{x}), \quad (15)$$

adding three more scalar constants (l, m, n) to the two Lamé constants. 1–2 week extension of `navier_solve_3d_tensor_mu`; design sketched in that file’s module docstring.

Grid. 32^3 vector Navier fits SuperLU on the login node; scaling to 48^3 (3 mm voxels matching the Yin acquisition) needs a memory-larger node ($\gtrsim 40$ GB) or an iterative solver with an appropriate preconditioner.

12 Files and reproducibility

All under `/u/sobh/Eman.Richard/tsm.fno/`:

File	Purpose
<code>src/solver/helmholtz_fd_3d.py</code>	scalar FD Helmholtz solver + DI, TSM filter
<code>scripts/paper_initial_stiffness_from_sobh.py</code>	build G_{P1}, G_{P2} from paper constants
<code>scripts/paper_wave_sim_sobh_initial.py</code>	bottom-plate + DI forward run
<code>scripts/paper_wave_sim_sobh_mip.py</code>	multi-face + TSM MIP forward run
<code>scripts/paper_wave_sim_sobh_vector.py</code>	vector Navier + rotated tensor μ_{ij}
<code>results/paper_initial_stiffness_sobh/</code>	$G(\mathbf{x})$ npz + PNGs + summary
<code>results/paper_wave_sim_sobh_initial/</code>	bottom-drive results
<code>results/paper_wave_sim_sobh_mip/</code>	TSM MIP results, shell stats, histograms
<code>results/paper_wave_sim_sobh_vector/</code>	vector Navier results

Python environment: `/u/sobh/.conda/envs/mri_mrf_pytorch_env` (numpy, scipy, matplotlib).
Runs take a few minutes on a login node; no GPU required.

Version: this document was regenerated from live scripts on September 15, 2026. Any change to the scripts is captured in git under `github.com:nahilsobh/Eman.Richard`; the latest relevant commits are `0bb5fb5`, `ade3bfc`, `dc8d796`, `a0fbafd`.

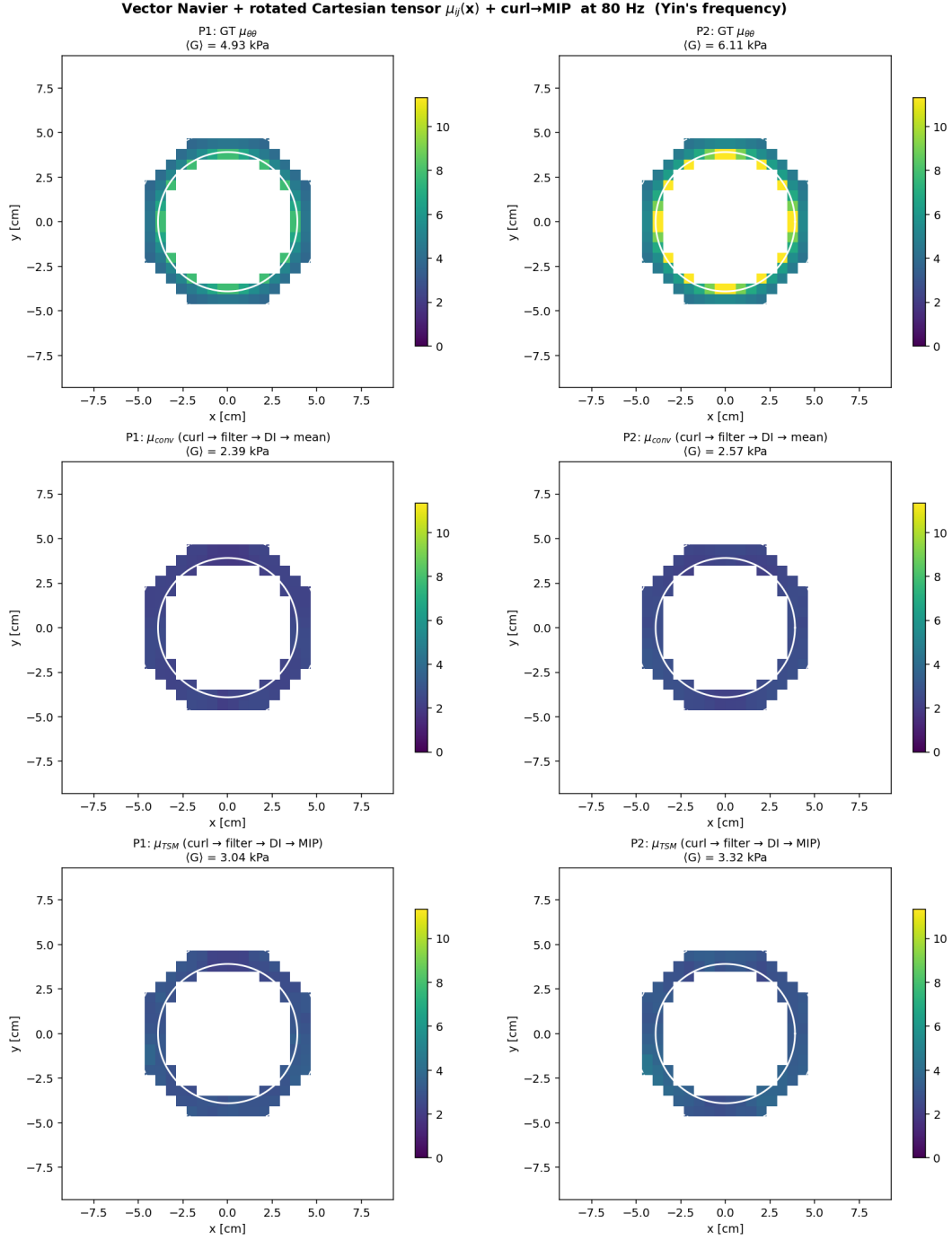


Figure 4: Shell heatmaps for the vector-Navier + tensor- μ_{ij} + curl→MIP pipeline at 80 Hz. Rows: GT $\mu_{\theta\theta}$ (top), μ_{conv} (middle), μ_{TSM} (bottom). Columns: P1 (left), P2 (right). All panels on the same colour scale. Grid is coarser (32^3 at 6 mm) than the scalar runs (60^3 at 3 mm) because vector Navier triples the DOF count.